

Comparing DeltaQuant with FreeAct

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Implementation Details For FreeAct Table 1

- Calibration data: 128 samples from WikiText2 with 2048 text tokens each followed by 256 mask tokens (maskcalib)
- Number of few shot examples: GSM8K (4-shot), Math500 (4-shot), HumanEval (0-shot)
- Other Hyperparameters: gen_length=256, block_length=32
- Evaluations done by lm-eval and accelerated by **Fast-dllm** (which I did not use previously)

	dLLMs			
	HumanEval	GSM8K	Math500	Avg.
	<i>LLaDA</i>			
16-bit Baseline	43.90	76.80	38.40	53.03
RTN (W8A8)	39.02	77.56	37.00	51.19
RTN (W4A4)	00.00	00.00	00.14	00.05
SmoothQuant	00.00	00.00	00.00	00.00
QuaRot	28.66	65.81	24.20	39.56
FlatQuant	39.63	74.37	33.60	49.20
FreeAct (ours)	42.68	75.74	34.60	51.01
	<i>Dream</i>			
16-bit Baseline	56.10	68.16	42.20	55.49
RTN (W8A8)	48.78	68.76	40.80	52.78
RTN (W4A4)	00.00	00.53	0.018	00.18
SmoothQuant	00.00	00.00	0.014	00.05
QuaRot	39.02	56.56	25.80	40.46
FlatQuant	46.34	63.53	34.60	48.16
FreeAct (ours)	50.00	68.84	32.60	50.48

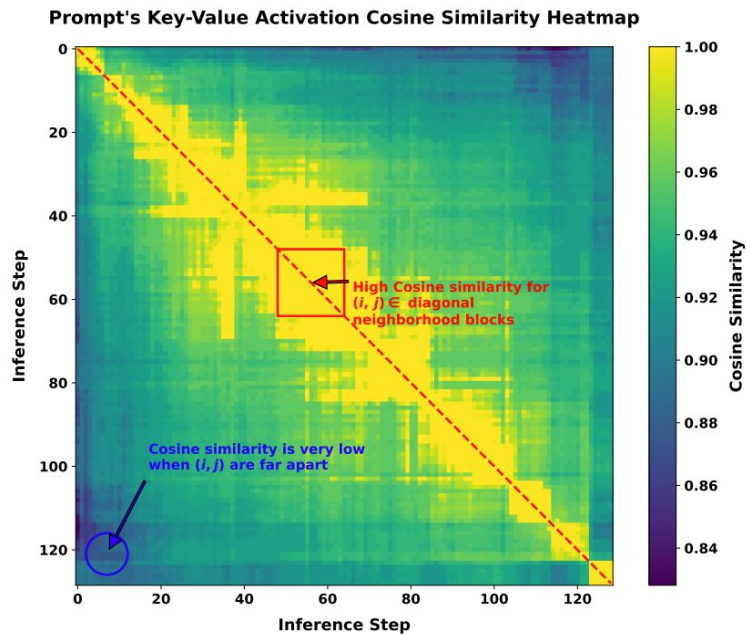
Fast-dllm

- A method that uses a block-wise approximate KV cache to speed up dllms inference.

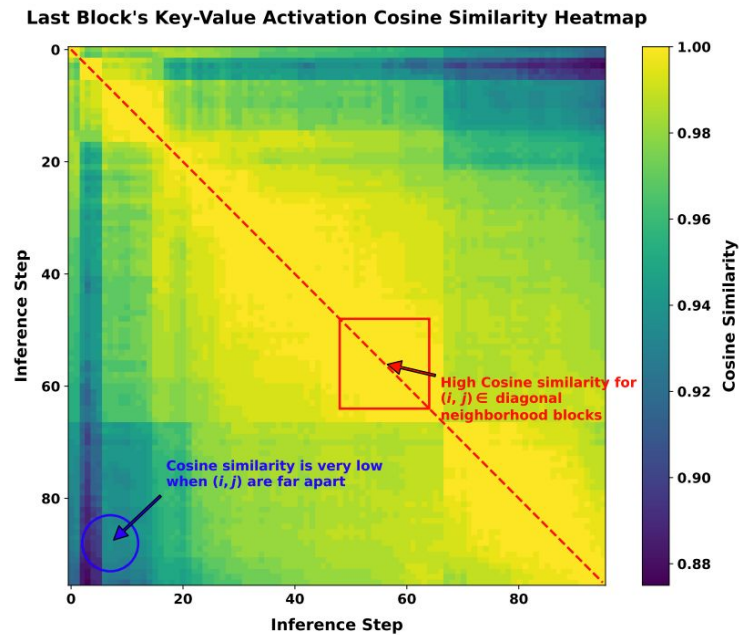
They proposed two types of caches:

- Prefix cache: before generating a new block, this method computes and stores the KV cache of other preceding blocks
- Dual cache: compute the KV cache for both prefix and suffix tokens

Fast-dllm

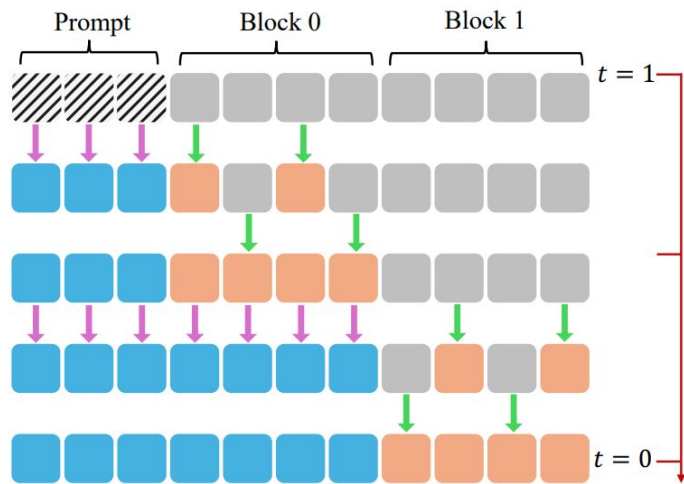


(a) Prompt block

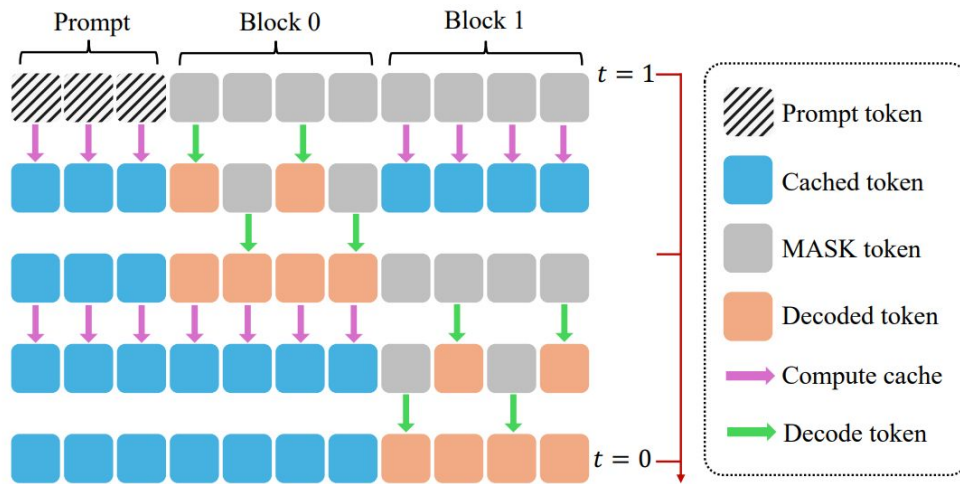


(b) Last block

Fast-dLLM



(a) Prefix KV Cache for block-wise generation.



(b) **DualCache**: Bidirectional KV cache contains prefix and suffix Cache.

FreeAct Implementation Possibilities

- They used Fast-dllm but did not specify which type of cache and if flash attention was used or not
- The possible combinations are:
 - No-cache
 - No cache + flash-attention
 - Prefix cache
 - Prefix cache + flash-attention
 - Dual cache
 - Dual cache + flash attention
- They also report results for QuaRot without enough implementation details about weight quantizer being used (RTN or GPTQ) and quantization symmetry (symmetric or asymmetric)

BF16 and QuaRot Comparisons

Table 17: Replication of FreeAct Table 1 Results (LLaDA-8B-Instruct) by finding the best hyperparameters and configurations

Config	HumanEval	GSM8K	Math500
FreeAct (paper) – 16-bit baseline	43.90	76.80	38.40
FreeAct (paper) – QuaRot W4A4	28.66	65.81	24.20
<i>Our 16-bit baseline (Fast-dLLM harness, LLaDA-8B-Instruct)</i>			
no cache (flash-attn)	41.46	78.09	38.0
no cache, SDPA	–	77.48	39.4
prefix cache (flash-attn)	43.29	77.10	38.4
prefix cache, SDPA	43.29	77.86	38.0
dual cache (flash-attn)	37.80	–	–
<i>Our QuaRot W4A4 (prefix cache, flash-attn)</i>			
GPTQ + asym (wikitext2, seqen 2048)	35.37	74.30	35.2
GPTQ + sym	33.54	74.07	31.8
RTN + asym	32.93	70.28	29.6
RTN + sym	32.32	68.23	28.4
GPTQ + maskcalib, asym	35.98	74.68	33.0
GPTQ + maskcalib, sym	36.59	73.31	31.2
<i>Our QuaRot W4A4 (no cache, flash-attn)</i>			
GPTQ + sym	34.76	74.68	35.2
RTN + asym	32.93	70.36	30.2
RTN + sym	28.05	69.29	29.0
GPTQ + maskcalib, asym	39.63	74.30	36.0
GPTQ + maskcalib, sym	33.54	73.84	35.6

Delta Quant Evaluation using Fast-dLLM (no cache)

LLaDA-8B-Instruct accuracy: our reproduction via Fast-dLLM (no-cache, real FlashAttention) vs. FreeAct’s own reported numbers, at W4A4.

Model	Setting	Method	HumanEval	GSM8K	MATH500	Avg.	$\Delta\%$
LLaDA-8B-Instruct	BF16	Base (ours, Fast-dLLM)	41.46	78.09	38.0	—	—
	BF16	Base (FreeAct paper)	43.90	76.80	38.40	53.03	—
	W4A4	QuaRot (ours, RTN+sym)	28.05	69.29	29.0	42.11	
	W4A4	QuaRot (FreeAct paper)	28.66	65.81	24.20	39.56	
	W4A4	DeltaQuaRot (1 anchor, ours)	35.37	—	33.6	—	—
	W4A4	DuQuant (maskcalib, ours)	36.59	—	38.0	—	—
	W4A4	DeltaDuQuant (1 anchor, ours)	37.80	—	34.6	—	—
	W4A4	FreeAct (paper’s own method)	42.68	75.74	34.60	51.01	↓3.81%

QuaRot Uniform Reanchoring Results

LLaDA-8B-Instruct accuracy: QuaRot-Delta uniform-reanchoring sweep (num_anchors $\in \{1, 2, 4, 8, 16, 32\}$).

Model	Setting	Method	Anchors	HumanEval	GSM8K	MATH500	Avg.	$\Delta\%$
LLaDA-8B-Instruct	BF16	Base (ours, Fast-dLLM)		41.46	78.09	38.0	52.52	—
		Base (FreeAct paper)		43.90	76.80	38.40	53.03	—
		QuaRot (ours, RTN+sym)		28.05	69.29	29.0	42.11	↓19.82%
	W4A4	QuaRot-Delta	1	35.37	—	33.6	—	—
			2	35.98	—	—	—	—
			4	37.80	—	—	—	—
			8	35.98	—	—	—	—
			16	39.02	—	—	—	—
			32	37.80	—	—	—	—
		FreeAct (paper’s own method)		42.68	75.74	34.60	51.01	↓3.81%

DuQuant Uniform Reanchoring Results

LLaDA-8B-Instruct accuracy: DuQuant-Delta uniform-reanchoring sweep (num_anchors $\in \{1, 2, 4, 8, 16, 32\}$).

Model	Setting	Method	Anchors	HumanEval	GSM8K	MATH500	Avg.	$\Delta\%$	
LLaDA-8B-Instruct	BF16	Base (ours, Fast-dLLM)		41.46	78.09	38.0	52.52	—	
		Base (FreeAct paper)		43.90	76.80	38.40	53.03	—	
		DuQuant (ours, maskcalib)		36.59	—	38.0	—	—	
	W4A4	DuQuant-Delta	1	37.80	—	34.6	—	—	
			2	37.20	—	—	—	—	
			4	40.24	—	—	—	—	
			8	39.02	—	—	—	—	
			16	39.63	—	—	—	—	
			32	42.07	—	—	—	—	
			FreeAct (paper’s own method)		42.68	75.74	34.60	51.01	↓3.81%

QuaRot Clipping Results

Table 4: LLaDA-8B-Instruct: QuaRot-Delta anchor & clip sweep.

Model	Setting	Method	Clip	Anchors	HumanEval	GSM8K	MATH500	Avg.	$\Delta\%$
LLaDA-8B-Instruct	BF16	Base (ours, Fast-dLLM)			41.46	78.09	38.0	52.52	—
		Base (FreeAct paper)			43.90	76.80	38.40	53.03	—
		QuaRot (ours, RTN+sym)			28.05	69.29	29.0	42.11	↓19.82%
	W4A4	QuaRot-Delta		1	35.37	—	33.6	—	—
				2	35.98	—	—	—	—
				4	37.80	—	—	—	—
				8	35.98	—	—	—	—
				16	39.02	—	—	—	—
				32	37.80	—	—	—	—
			0.99	2	34.15	—	—	—	—
				4	34.76	—	—	—	—
				8	35.37	—	—	—	—
				16	35.98	—	—	—	—
				32	33.54	—	—	—	—
			0.999	2	33.54	—	—	—	—
				4	34.76	—	—	—	—
				8	35.98	—	—	—	—
				16	33.54	—	—	—	—
				32	34.15	—	—	—	—
			0.9999	2	35.98	—	—	—	—
				4	37.20	—	—	—	—
				8	35.98	—	—	—	—
				16	35.37	—	—	—	—
				32	35.37	—	—	—	—
		FreeAct (paper’s own method)			42.68	75.74	34.60	51.01	↓3.81%

DuQuant Clipping Results

Table 5: LLaDA-8B-Instruct: DuQuant-Delta anchor & clip sweep.

Model	Setting	Method	Clip	Anchors	HumanEval	GSM8K	MATH500	Avg.	$\Delta\%$	
LLaDA-8B-Instruct	BF16	Base (ours, Fast-dLLM)			41.46	78.09	38.0	52.52	—	
		Base (FreeAct paper)			43.90	76.80	38.40	53.03	—	
		DuQuant (ours, maskcalib)			36.59	—	38.0	—	—	
	W4A4	DuQuant-Delta		1	37.80	—	34.6	—	—	
			2	37.20	—	—	—	—		
			4	40.24	—	—	—	—		
			8	39.02	—	—	—	—		
			16	39.63	—	—	—	—		
			32	42.07	—	—	—	—		
			0.99	2	40.24	—	—	—	—	
				4	35.37	—	—	—	—	
				8	38.41	—	—	—	—	
				16	41.46	—	—	—	—	
				32	42.68	—	—	—	—	
				0.999	2	40.85	—	—	—	—
			4		40.85	—	—	—	—	
			8		39.02	—	—	—	—	
			16		37.80	—	—	—	—	
			32		39.02	—	—	—	—	
			0.9999		2	38.41	—	—	—	—
				4	39.02	—	—	—	—	
				8	38.41	—	—	—	—	
				16	39.02	—	—	—	—	
				32	39.02	—	—	—	—	
				FreeAct (paper’s own method)				42.68	75.74	34.60

Delta Quant Evaluation using Prefix Cache

LLaDA-8B-Instruct accuracy: our reproduction via Fast-dLLM (prefix cache, real FlashAttention) vs. FreeAct’s own reported numbers, at W4A4.

Model	Setting	Method	HumanEval	GSM8K	MATH500	Avg.	$\Delta\%$
LLaDA-8B-Instruct	BF16	Base (ours, Fast-dLLM)	43.29	77.10	38.4	52.93	—
	BF16	Base (FreeAct paper)	43.90	76.80	38.40	53.03	—
	W4A4	QuaRot (ours, RTN+sym)	32.32	68.23	28.4	42.98	↓18.80%
	W4A4	QuaRot (FreeAct paper)	28.66	65.81	24.20	39.56	↓25.40%
	W4A4	DeltaQuaRot (16 anchors, ours)	37.80	—	32.4	—	—
	W4A4	DuQuant (maskcalib, ours)	36.59	—	36.4	—	—
	W4A4	DeltaDuQuant (16 anchors, ours)	40.85	—	35.2	—	—
	W4A4	FreeAct (paper’s own method)	42.68	75.74	34.60	51.01	↓3.81%

Reproducing DLLMQuant Results

Table 8:LLaDA-8B-Base accuracy: our reproduction vs. DLLMQuant’s own reported numbers, at W4A4.

Model	Setting	Method	HumanEval	GSM8K	Avg. [§]
LLaDA-8B-Base	FP16	Base (DLLMQuant paper)	32.92	67.48	50.20
	FP16 (gen_len 256)	Base (ours)	32.93	66.64	49.79
	FP16 (gen_len 512)	Base (ours)	32.32	–	–
	FP16 (gen_len 1024)	Base (ours)	31.10	–	–
	W4A4	QuaRot GPTQ (ours, WikiText2 calib, sym)	23.78	53.30	38.54
	W4A4	QuaRot GPTQ (ours, WinoGrande calib, asym) [‡]	26.22	–	–
	W4A4	QuaRot (DLLMQuant paper, WinoGrande calib)	25.33	44.57	34.95
	W4A4	DLLMQuant++ (paper’s own method, QuaRot-based)	28.92	56.25	42.59

Plan for Next Week

- Report missing results from the tables for other benchmarks
- Conduct experiments for Dream model
- Compare with QuantDLLM
- Do experiments with MACaQ quantization